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FitQT: a fully integrated and trainable quantum transformer for scalable natural language processing

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Abstract

The combination of Transformer architectures with Quantum Computing has opened new directions in Quantum Natural Language Processing (QNLP) by leveraging the Transformer's representational capacity together with the computational parallelism and high-dimensional feature space of quantum systems. However, existing approaches remain constrained by partial designs and limited integration, hindering scalability and trainability. This study introduces FitQT—a novel, fully integrated, and trainable quantum transformer that unifies three essential components: (1) a quantum multi-head attention mechanism employing a scaled Gaussian kernel, (2) a quantum feed-forward network, and (3) quantum residual connections. The residual connections stabilize gradient flow and facilitate the training of deeper quantum circuits, while the attention and feed-forward components jointly enhance representational capacity. Evaluations on sentiment analysis datasets demonstrate FitQT's effectiveness. On the IMDb dataset, FitQT improves accuracy by 14.89% over the baseline Quantum Self-Attention Network and achieves competitive task-level performance relative to BERT-Base, despite being trained from scratch without large-scale pretraining, while requiring only 2,595 parameters and substantially reduced training time. Moreover, FitQT achieves state-of-the-art performance compared to prior quantum-based transformer architectures such as QSANN and QMSAN. These results highlight the importance of incorporating classical deep learning principles into quantum architectures and provide a practical foundation for scalable, resource-efficient QNLP systems suitable for the noisy intermediate-scale quantum era.

1. Introduction

Quantum computing has demonstrated provable computational speedups for specific problems such as integer factorization and unstructured search, and continues to show promise in optimization, secure communication, chemical engineering, and machine learning [1–7]. Recent advances in quantum algorithms and hardware in both academia and industry are accelerating the practical development of quantum technologies. In particular, progress in quantum hardware, including IBM's 1121-qubit superconducting processor [8], D-Wave's 5000-qubit quantum annealer [9], and neutral-atom quantum systems with 280 physical qubits [10], is providing a foundation for the deployment of advanced quantum algorithms in life sciences and artificial intelligence.

Quantum algorithms have been proposed for complex calculations, including ground state computations [11, 12], solving linear systems [13], and predicting drug-receptor interactions [14] as well as protein folding [15, 16], which are major challenges in molecular biology and drug discovery. Particularly, quantum machine learning is being explored for potential applications in synthetic chemistry and biology, with researchers investigating whether quantum approaches could offer advantages for generative chemistry tasks [15, 17]. The combination of AI, quantum computing, and the physics of complex systems is set to solve challenges related to scale and data complexity in the life sciences. This integration enhances the accuracy and generalizability of models, leading to significant advancements in both research and practical applications.

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A new multidisciplinary area called Quantum Natural Language Processing (QNLP) is developing, which combines the strengths of quantum computing and NLP. Initial explorations in this domain have investigated quantum-inspired models for various linguistic tasks, including information retrieval [18, 19], logic-based disambiguation [20], and compositional semantics [21, 22]. More recently, the execution of fundamental NLP algorithms on quantum hardware has transitioned from theoretical possibility to practical feasibility, with early implementations yielding promising outcomes [23, 24]. Current research endeavors are increasingly focused on the quantum modeling of key linguistic components such as embeddings, sequential processing, attention mechanisms, and grammatical structures, with the overarching goal of developing scalable QNLP frameworks.

Given that the attention mechanism is a cornerstone of state-of-the-art classical NLP and a natural candidate for quantum adaptation, it is essential to consider the model that established its prominence. The Transformer has become a central architecture in Artificial Intelligence (AI) and serves as the foundation of modern generative systems [25]. At its core lies the attention mechanism, originally designed to model dependencies among language tokens [26]. The seminal work 'Attention Is All You Need' [27] introduced the Transformer to address the limitations of Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs), while formalizing key components such as self-attention, multi-head attention, and word embeddings. Built upon an encoder–decoder structure [28], the model stacks self-attention and fully connected layers [29]. Its extensive use of residual connections plays a critical role in stabilizing training and enabling deeper architectures. Initially developed for NLP tasks including machine translation, content generation, and sentiment analysis [30], the Transformer has since become a dominant framework across AI.

Its adaptability has enabled successful applications beyond NLP. In computer vision [25, 31, 32], Vision Transformers demonstrate strong capabilities in image representation and salient region detection [33]. Transformer-based models have also been applied to tabular data, such as the Feature Tokenization Transformer (FTT) [34]. This broad applicability has established the Transformer as a standard architecture for contemporary AI systems. Building upon classical pre-training paradigms, QNLP research has begun integrating self-supervised learning strategies. Inspired by models such as BERT, a quantum-enhanced BERT with a unitary classification head has achieved performance comparable to its classical counterpart on the GLUE benchmark [35]. These results indicate the feasibility of hybrid quantum–classical architectures for advanced language understanding tasks and motivate further exploration within QNLP.

Recent years have seen growing interest in quantum-enhanced transformer architectures. Researchers are exploring whether the unique properties of quantum systems, particularly their high-dimensional Hilbert spaces and entanglement capabilities, might provide computational advantages for certain classes of learning problems. However, most efforts to date have focused on vision benchmarks rather than language tasks. For example, quantum transformers have been applied to MNIST classification using quantum state amplitude encoding [36], while modular quantum adaptations of transformer components remain largely theoretical [37, 38]. Grover's algorithm has been incorporated into quantum transformer frameworks for Fashion MNIST [39], and sparse attention mechanisms inspired by Grover's search have been proposed [40], though practical implementations are limited. A quantum kernel-based transformer was also tested for binary classification on MNIST and Fashion MNIST [41]. While these studies demonstrate growing interest in quantum transformers, they highlight a significant gap in their applicability to large-scale NLP tasks.

While the application of quantum computing to vision-based benchmarks has seen notable exploration, its potential within Natural Language Processing (NLP) through Quantum Transformers (QTs) remains a compelling yet significantly underexplored frontier [7, 42]. Early forays into adapting Transformer components for the quantum domain, such as preliminary quantum self-attention mechanisms [43–45], have been reported. However, these pioneering efforts are often fundamentally incomplete, typically restricted to single-head attention and critically omitting other essential Transformer building blocks. Specifically, they often lack robust implementations of quantum multi-head attention, crucial for capturing diverse contextual dependencies from multiple perspectives, and quantum feed-forward networks, vital for introducing non-linear transformations and enhancing model expressiveness. This fragmentation of components hinders the realization of a truly holistic and powerful quantum Transformer architecture capable of tackling complex NLP tasks.

The architectural strengths of Transformers, particularly their parallel processing capabilities, make them a more promising candidate for quantum adaptation compared to sequential models like quantum recurrent neural networks (QRNNs) [46, 47]. As quantum hardware continues its progression towards error-corrected, fault-tolerant systems [48, 49], and as the theory of quantum algorithms continues to develop [50], the prospect of executing more comprehensive quantum-enhanced Transformers becomes increasingly tangible. However, whether such systems can demonstrate practical advantages over classical transformers for language tasks remains an important open question.

Despite this uncertainty, the evolving QNLP landscape underscores a pressing need: the development of a fully integrated quantum Transformer architecture. Such an architecture must not only incorporate all key functional modules emulating its classical counterpart but also be meticulously designed for robust trainability

and scalability, especially considering the constraints of near-term and intermediate-scale quantum (NISQ) devices. However, constructing such a comprehensive, multi-component quantum architecture invariably introduces significant challenges related to circuit depth, parameter optimization, and information fidelity.

To overcome these inherent challenges and enable a complete quantum Transformer to be effectively trained and scaled to the necessary depth for sophisticated NLP applications, it must integrate critical design principles proven effective in deep classical networks. Foremost among these is the incorporation of effective quantum residual connections. These mechanisms are indispensable for stabilizing training dynamics in deep quantum circuits, facilitating robust information propagation across numerous layers, and critically, mitigating deleterious effects such as information degradation due to quantum noise and the pervasive barren plateau phenomenon, which can render deep quantum models practically untrainable. While the development of individual quantum components like attention and embeddings is progressing, their assembly into a deep, functional, and trainable Quantum Transformer architecture is contingent upon addressing these fundamental issues of information flow and optimization stability. Thus, a cohesive, end-to-end quantum Transformer must be not only structurally complete but also architecturally sound, leveraging mechanisms like residual connections to ensure its viability and performance.

Acknowledging the critical gaps and the urgent imperative previously outlined, this paper introduces an in-depth investigation into a novel Quantum Transformer architecture, meticulously designed to address the current limitations in Quantum Natural Language Processing (QNLP). Our research places a strong emphasis on constructing a holistic and functionally complete Quantum Transformer framework, with a particular focus on the robust integration and synergistic operation of three pivotal components: quantum multi-head attention mechanisms, quantum residual connections, and quantum feed-forward networks. We posit that the cohesive assembly of these elements is fundamental to systematically investigating the feasibility, trainability, and empirical behavior of deep quantum learning architectures for language tasks, without presupposing any inherent quantum advantage over classical counterparts.

To this end, this study represents the next phase of our ongoing research on quantum transformer models, following our earlier work presented at ICIT 2025, titled 'QPMHA: Gaussian Projected Quantum Multi-Head Attention for Sentiment Analysis' [51]. The primary contribution of this work is the design of a novel and comprehensive Quantum Transformer architecture that integrates quantum multi-head attention to capture diverse contextual relationships, quantum position-wise feed-forward networks to enhance non-linear representational ability, and quantum residual connections to ensure stable training and effective information propagation through deep quantum circuits. Building on this architecture, we empirically evaluate the performance of the proposed model on the sentiment analysis task in both English and Vietnamese, comparing it against both quantum and classical transformer models to analyze training dynamics, scalability, and overall effectiveness. Finally, we provide practical insights into constructing scalable quantum-enhanced NLP systems, highlighting the advantages and considerations for leveraging near-term and future quantum computational resources.

2. Preliminaries and related work

2.1. Original transformer architecture

An in-depth analysis of the architecture and implementation outlined in [27] is essential for a comprehensive understanding and investigation of the factors that lead to the success of the Transformer model. The Transformer preserves the encoder–decoder structure, as seen in figure 1, but removes recurrence and convolution components, in contrast to previously suggested sequence transduction models such as [52] and [53]. Besides the point-wise feedforward networks included in the Transformer model, an innovative multi-head attention mechanism is introduced to enable effective sequence modeling. The general Transformer architecture put forth by [27] is depicted in figure 1. The building blocks of this architecture are discussed in detail in the following sections.

2.2. Multi-head attention

The attention mechanism lies at the core of the Transformer architecture [27], enabling the computation of word (or token) representations by dynamically focusing on important parts in the sequence, as illustrated in figure 2. Furthermore, this computation can be performed in parallel, unlike the sequential processing required by recurrent neural networks.

The most fundamental form used in Transformers is the Scaled Dot-Product Attention, shown in equation (1):

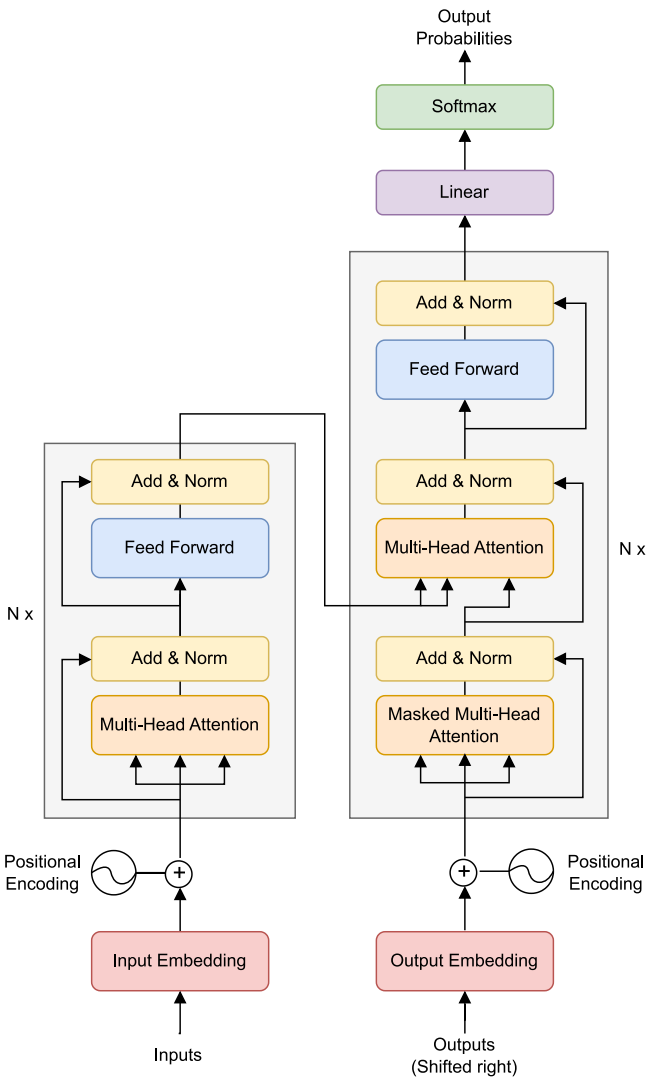


Figure 1. The Transformer architecture [27].

$$\text{Attention}(Q,\ K,\ V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V, \tag{1}$$

where Q, K, V represent the matrices of queries, keys, and values, respectively, and d_k denotes the dimension of the keys. The attention scores are computed by taking the dot product between the query and key vectors, $S = QK^{\top}$. The magnitude of the raw dot product may become large, causing the output of the softmax function to be very small and resulting in vanishing gradients. To mitigate this, the raw dot product is scaled by $\frac{1}{\sqrt{d_k}}$. The scaled scores are then processed by the softmax function and subsequently multiplied by the value matrix V. These resulting weights indicate how much each word is influenced by all other words in the sequence.

Building on this, Multi-Head Attention enables the model to compute multiple representations of the sequence in parallel, each capturing different aspects of the relationships between words. Mathematically, it is formed as:

$$\text{MultiHead(Q, K, V)} = \text{Concat}(\text{head}_1,\ \text{head}_2, \ldots, \text{head}_h)\, W^O, \tag{2}$$

where each attention head is computed as

$$\text{head}_i = \text{Attention}(QW_i^Q,\ KW_i^K,\ VW_i^V),$$

with W_i^Q, W_i^K, and W_i^V indicating learnable projection matrices for the ith head, and W^O is the output projection matrix.

By computing attention across multiple heads simultaneously, syntactic and semantic relationships can be captured, which are then aggregated to form a richer, multi-faceted representation of the input sequence.

2.2.1. Feed-forward networks

The feedforward network, positioned subsequent to the self-attention layers in both the encoder and decoder, is an essential component in the functionality of transformers. This network has two linear transformations and a non-linear ReLU activation function, executed separately and identically at each point. The model can autonomously learn intricate changes at every point thanks to this architectural design, which guarantees that every token is handled consistently.

The feed-forward network enhances the representation learned from the attention mechanism by adding additional layers of abstraction and complexity to the token embeddings. Its role is to refine the contextualized representations and capture intricate patterns within the data. The mathematical model for this element is presented in equation (3):

$$\mathrm{FFN}(x) = \mathrm{ReLU}(xW_1 + b_1)W_2 + b_2, \tag{3}$$

where W_1, W_2 represent the learnable transformation matrices, and b_1, b_2 are bias terms.

2.2.2. Residual connections

Additionally, the transformer applies layer-by-layer normalization [54] after implementing residual connections [55] around each module. By enabling gradients to flow directly across layers, residual connections help alleviate the vanishing gradient issue. Therefore, we may use the formulation shown in equation (4) to represent each transformer block:

$$H' = \mathrm{SelfAttention}(X) + X, \quad H = \mathrm{FFN}(H') + H' \tag{4}$$

Residual connections improve model performance by facilitating more effective information flow across layers. This equation's '+' operator, which stands for element-wise addition, aids in resolving the vanishing gradient issue.

2.3. Quantum computing concepts

2.3.1. Qubit

In quantum computing, a qubit, short for quantum bit, serves as the fundamental unit of information, comparable to the bit in classical computing. In contrast to a traditional bit, which can represent only the values 0 or 1, a qubit can exist in a superposition of both basis states $|0\rangle$and$|1\rangle$. The state of a qubit is explicitly represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{5}$$

where α and β are complex numbers that adhere to the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1 \tag{6}$$

Due to the laws of superposition and quantum entanglement, qubits can encode and process information in ways unattainable by classical systems. This presents the possibility for quantum algorithms to exceed classical counterparts in speed and efficiency when addressing intricate issues in fields such as artificial intelligence, cryptography, and optimization.

2.3.2. Entanglement

Quantum entanglement is a distinctive phenomenon in quantum mechanics, wherein the states of several qubits become profoundly coupled, making an independent description impossible. When qubits are entangled, the state of each qubit depends on the overall system, meaning that measuring one qubit instantly affects the others, mregardless of the spatial separation between them.

Initially a controversial theoretical concept, entanglement has now been experimentally confirmed and serves as a fundamental component in modern quantum computer architectures. Entangling gates, such as the CNOT gate, are used to establish these correlations between qubits and are essential for quantum information processing.

Remarkably, a single type of two-qubit entangling gate, when combined with single-qubit gates, is enough to make up a universal gate set, facilitating any quantum computation. Entanglement is what gives quantum computers their edge: it allows them to perform calculations that would require enormous resources on classical computers, due to the exponential growth in the complexity of entangled quantum systems.

2.3.3. Variational quantum circuits

(VQCs) have become a fundamental component in the Noisy Intermediate-Scale Quantum (NISQ) computing era, offering a promising method to leverage quantum computers for intricate challenges, especially in artificial intelligence [56, 57]. In essence, VQCs represent a class of hybrid quantum–classical algorithms wherein a quantum circuit with adjustable parameterized gates is optimized via a feedback loop with a classical processor.

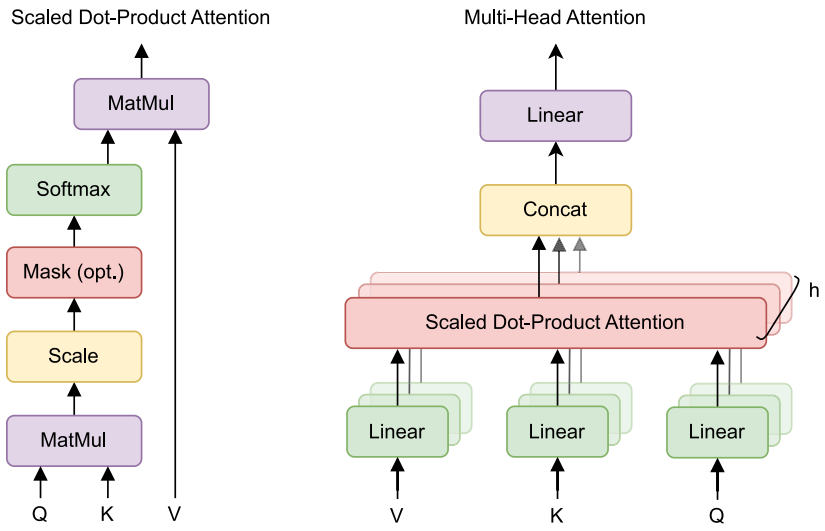


Figure 2. Scaled dot-product attention and multi-head attention in the transformer model [27].

This architecture allows VQCs to utilize the extensive high-dimensional Hilbert space of quantum systems to investigate intricate solution spaces, while reducing circuit depth requirements and fault-tolerance needs, in accordance with the constraints of existing quantum hardware. The structural and conceptual analogy to classical neural networks, where parameters are trained to perform specific tasks, has spurred the application of VQCs as fundamental building blocks for Quantum Machine Learning models [6, 58, 59]. The inherent expressiveness and entangling capacity of VQCs offer the theoretical ability to achieve potential quantum advantages over classical systems in modeling intricate correlations and high-dimensional data patterns, thereby promoting progress in areas such as optimization, quantum chemistry, pattern recognition, and natural language processing.

Building upon this foundation, the application of VQCs to NLP, often termed Quantum NLP, is an actively developing research frontier. In QNLP, VQCs are ingeniously adapted to model the complex semantic and syntactic structures inherent in human language. This often involves designing VQC-based modules that act as quantum counterparts to components of successful classical deep learning architectures, such as quantum recurrent units for processing sequential data like sentences [60]. Furthermore, VQCs are employed to implement compositional models of meaning, where the semantics of larger linguistic units are derived from their constituents, often drawing inspiration from categorical grammar frameworks [61]. These quantum circuits are subsequently trained, typically using hybrid quantum–classical optimization loops, to perform various NLP tasks, such as classification of text and sentiment analysis, demonstrating the ability of VQCs to derive meaningful representations from linguistic input [62].

2.3.4. Measurement and expectation value

The transformation of information from quantum state space to an interpretable classical form is achieved through measurement, a crucial interface in quantum information processing [63–65]. In Quantum Machine Learning, the classical data generated by this measurement is essential for training and assessing machine learning models. In contrast to unitary quantum operations that are reversible, measurement is fundamentally probabilistic and irreversible, resulting in the collapse of the quantum state. Central to this process is the concept of the expectation value, a statistical quantity that provides key information from the quantum state for training and evaluating Quantum Machine Learning (QML) models. Consider a quantum system, typically encoded in a register of n qubits, described by a normalized state vector $|\psi\rangle$ belonging to the Hilbert space $H = (\mathbb{C}^2)^{\otimes n}$. A physical observable, representing a feature or an objective function that the QML model needs to estimate, is represented by a Hermitian operator A acting on H. The expectation value of A with respect to the state $|\psi\rangle$, represented as $\langle A\rangle$, is precisely specified by the inner product expression:

$$\langle A\rangle = \langle\psi|A|\psi\rangle \tag{7}$$

In expression (7), the operator A acts on the state vector $|\psi\rangle$, and the resulting vector (another vector in H) is then projected onto the vector $|\psi\rangle$ (the bra corresponding to $|\psi\rangle$). Physically, the expectation value $\langle A\rangle$ represents the statistical average of the possible outcomes obtained if the observable A were measured on a large ensemble of quantum systems prepared in the identical state $|\psi\rangle$.

In many QML applications, particularly in VQAs, the state $|\psi\rangle$ is often parameterized, $|\psi(\theta)\rangle$, by a set of classical parameters θ. In such cases, the expectation value $\langle A\rangle(\theta) = \langle\psi(\theta)|A|\psi(\theta)\rangle$ becomes a function of these parameters and frequently serves as the loss function, or a component thereof, which needs to be optimized during the model's training process.

If the operator A possesses a discrete spectrum with real eigenvalues $\{a_j\}$ and corresponding orthonormal eigenvectors $\{\ |\phi_j\rangle\ \}$ (such that $A|\phi_j\rangle = a_j|\phi_j\rangle$ and $\langle\phi_j|\phi_k\rangle = \delta_{jk}$), the expectation value can be equivalently expressed via Born's rule [66], highlighting the probabilistic nature of quantum measurement:

$$\langle A\rangle = \sum_j a_j\ |\langle\phi_j|\psi\rangle|^2 \tag{8}$$

Here, a_j are the possible measurement outcomes (eigenvalues of A), and $P(a_j) = |\langle\phi_j|\psi\rangle|^2$ is the probability of the measurement on state $|\psi\rangle$ yielding the outcome a_j. Consequently, formula (8) illustrates that the expectation value is exactly the weighted average of the eigenvalues, with the weights corresponding to their relative probabilities of occurrence. In QML models, a_j can correspond to class labels (in classification problems) or specific values, and the model is trained to adjust $|\psi(\theta)\rangle$ such that the probabilities $P(a_j)$ (and consequently $\langle A\rangle(\theta)$) accurately reflect the relationships within the data.

For instance, in a binary classification task using a single output qubit, the measurement operator could be the Pauli-Z operator, with eigenvalues $+1$ (corresponding to class 0) and -1 (corresponding to class 1). The expectation value $\langle Z\rangle$ would provide a continuous value in the range $[-1, 1]$, which can be directly used to make a classification decision. Estimating the expectation value in practice on a quantum computer requires performing the measurement multiple times and averaging the results. The precision of this estimate depends on the number of shots and is a critical factor to consider in QML algorithm design, especially when dealing with noise and the limitations of current quantum hardware [67, 68]. Accordingly, methodologies aimed at reducing the necessary number of measurements or enhancing the accuracy of the expectation value estimate are prominent research domains in Quantum Machine Learning.

2.4. Recent advances in quantum natural language processing architectures

Having introduced the fundamental concepts of quantum computing and Variational Quantum Circuits, we now turn to the emerging field of Quantum Natural Language Processing (QNLP). Inspired by the success of classical deep learning, particularly attention mechanisms and Transformer architectures, recent research has focused on developing quantum counterparts for NLP tasks. This section reviews recent advances, including Quantum Self-Attention mechanisms and more comprehensive Quantum Transformer architectures, and discusses the key challenges that continue to motivate further investigation.

2.4.1. Quantum self-attention mechanisms

The development of quantum self-attention mechanisms is an active area of research, with several distinct approaches being explored to adapt the principles of self-attention for quantum computation. These methods aim to leverage quantum phenomena for tasks such as text classification and image processing.

One line of research focuses on Quantum Natural Language Processing. For instance, a Quantum Self-Attention Neural Network (QSANN) [43] has been introduced for text classification. This method utilizes a Quantum Self-Attention Layer (QSAL) where classical input words are encoded into multi-qubit quantum states. Three distinct quantum ansatzes (U_q, U_k, U_v) are then applied to generate Query, Key, and Value representations. The Gaussian Projected Quantum Self-Attention (GPQSA) mechanism is employed, which Query and Key representations (specifically, the expectation values of the Pauli-Z operator, $\langle Z_q\rangle_s$ and $\langle Z_k\rangle_j$ are projected into classical values. The self-attention coefficients between words are subsequently calculated using a Gaussian function based on these projected classical values. The Quantum Self-Attention Neural Network comprises stacked Quantum Self-Attention Layers, followed by a classical processing layer. This approach is designed with shallow circuits suitable for implementation on Noisy Intermediate-Scale Quantum (NISQ) devices, employing the quantum Hilbert space to identify correlations between words.

Another approach, the Quantum Kernel Self-Attention Network [44], combines elements of Quantum Kernel Methods with the Self-Attention Mechanism. The Quantum Kernel Self-Attention Network utilizes a Quantum Kernel Self-Attention Mechanism: classical input data is encoded into quantum states. From these, Quantum Query States and Quantum Key States are formed. The Quantum Kernel Self-Attention Score, defined as $|\langle Q|K\rangle|^2$, is then computed. Concurrently, a Quantum Value State is prepared. This method applies a 'mathematical link' based on the Deferred Measurement Principle and conditional measurement techniques: the measured Quantum Kernel Self-Attention Score from a first quantum register controls a unitary operation applied to the Quantum Value State in a second quantum register. The Quantum Kernel Self-Attention Network architecture operates as a hybrid quantum–classical model. The application of the Deferred Measurement Principle and conditional measurements in the Quantum Kernel Self-Attention Mechanism is described,

with the objective of reducing quantum resource requirements through mid-circuit measurement, thereby influencing the method's feasibility and adaptability. The Quantum Kernel Self-Attention Score is generated with an 'exponentially large characteristic space' to accommodate information and determine measurement conditions.

Further research in quantum self-attention, particularly for image classification, has led to a Quantum Self-Attention Network [45] that employs a Quantum Self-Attention Mechanism. In this framework, quantum Query, Key, and Value states are generated. A key aspect is Quantum Logic Similarity, which uses quantum logic operations (instead of inner products) to compute the similarity between Query and Key states, with the result being a quantum state. This leads to the formation of a Quantum Bit Self-Attention Score Matrix, a density matrix with Quantum Logic Similarity states as its elements, representing the attention distribution. This method incorporates a 'one-step' circuit design, 'measurement compression' techniques, and the concept of 'Quantum Coordinates' to describe and optimize the circuits. The application of Quantum Logic Similarity and the Quantum Bit Self-Attention Score Matrix within the Quantum Self-Attention Mechanism allows for the entire self-attention process to be performed on a quantum computer without intermediate measurements, aiming to preserve quantum information. The 'Quantum Coordinates' design and optimization techniques like measurement compression are described, with the goal of enhancing computational efficiency and reducing resource demands on near-term quantum devices. The Quantum Bit Self-Attention Score Matrix is characterized by a higher information representation capacity compared to classical self-attention score matrices.

Research [69] introduces the Quantum Mixed-State Self-Attention Network (QMSAN), a hybrid quantum–classical neural network architecture designed to enhance the effectiveness of the self-attention mechanism in NLP tasks. The core methodology of QMSAN comprises two primary innovations. Firstly, the model proposes an attention similarity computation mechanism based on representing the Query and Key components as quantum mixed states. These mixed states are generated from initial pure states via partial trace operations over a subsystem of the quantum system, allowing for the retention of information about potential combinations and entanglement that pure states might not fully represent. Subsequently, the similarity between the two mixed Query and Key states, quantified by the value trace ($\rho_{\text{query}}\,\sigma_{\text{key}}$), is estimated directly within the quantum domain using a Swap Test quantum circuit. The use of the Swap Test enables direct computation of the overlap between the two density matrices without requiring individual measurements on each state followed by similarity computation on classical data, thereby preserving quantum information and potentially capturing more subtle correlations. Secondly, QMSAN introduces an advanced quantum positional encoding scheme, integrated into the main processing circuit without requiring additional auxiliary qubit resources. The positional information of tokens in a sequence is encoded via fixed quantum gate operations, specifically designed and embedded into the existing circuit architecture. A key feature of this scheme is the application of distinct scaling strategies for input data and positional information: positional information, with its natural periodicity, is scaled to utilize the full $[0, 2\pi]$ cycle of quantum rotation gates (e.g., Rz gates). Conversely, input data, which typically lacks intrinsic periodicity, is scaled to a more restricted range (e.g., $[0, \pi]$) to avoid inadvertently introducing unintended symmetries in the quantum representation space that could distort the original data's nature. Furthermore, QMSAN's quantum embedding architecture is designed with a repetitive structure, where data encoding blocks and trainable circuit blocks are interleaved, allowing learnable parameters to participate more deeply in the formation of the quantum feature space. These methodological innovations in QMSAN aim to improve the accuracy of attention coefficient computation and the effective integration of positional information, while also considering the resource limitations of current near-term quantum devices.

2.4.2. Quantum transformer

The core architectural innovation of Quixer [70] lies in the introduction of a novel attention mechanism based on quantum principles, replacing the dot-product-based self-attention mechanism found in classical Transformer architectures. Instead of representing tokens as vectors, Quixer encodes each token in the input sequence into a trainable unitary operator. It then employs the 'Linear Combination of Unitaries' technique to create a single composite operator, which is a weighted superposition of all token operators within the context window. To introduce complex non-linear interactions between tokens, Quixer further applies a non-linear transformation to this composite operator via the 'Quantum Singular Value Transform', which allows for the application of a trainable polynomial function to the operator. This combination of Linear Combination of Unitaries and Quantum Singular Value Transform creates a powerful method for mixing information, where interactions between tokens are not computed pairwise but are instead realized through the composition of quantum operators, enabling the model to capture more complex relationships, such as interactions between skip-k-grams, depending on the degree of the polynomial used. When evaluated on the language modeling task on the Penn Treebank dataset, the model demonstrated superior performance over LSTM architectures and

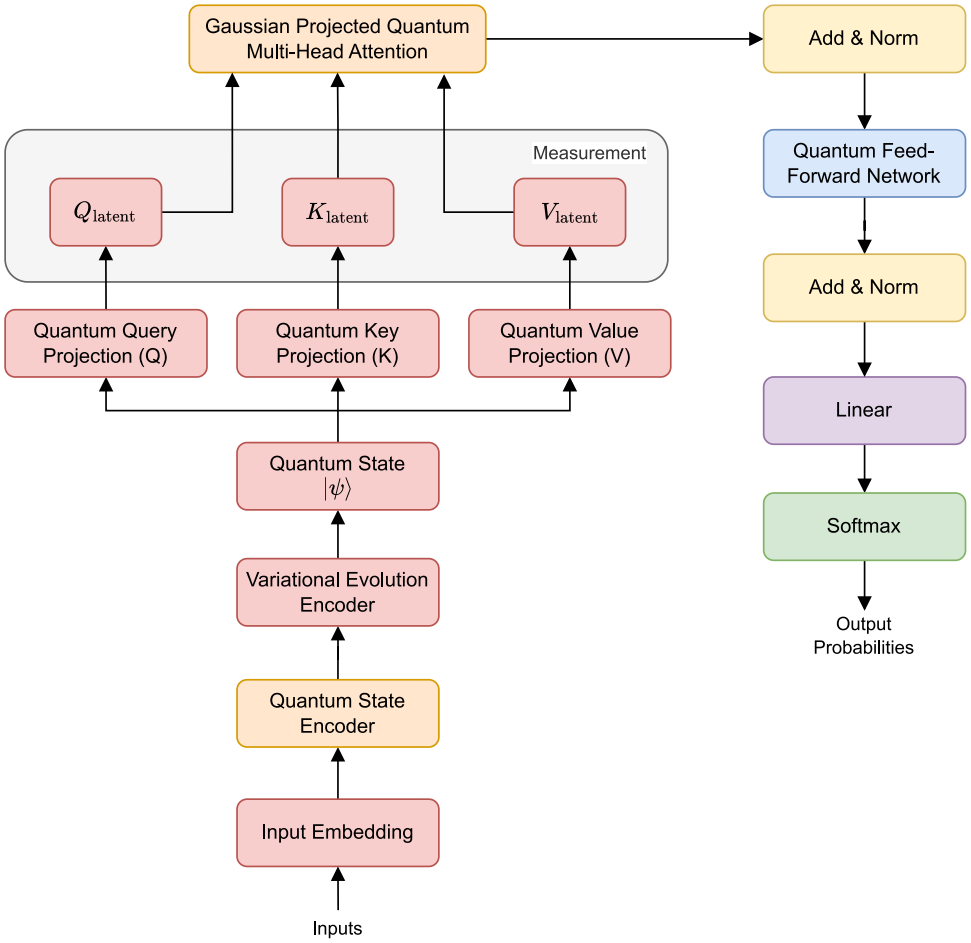


Figure 3. The overall architecture of the proposed FitQT model. The model processes input data through an embedding layer, followed by a Quantum State Encoder and a Variational Evolution Encoder to prepare a quantum state representation, $|\psi\rangle$. This state is then fed into three parallel quantum projection circuits to generate Query (Q), Key (K), and Value (V). The detailed circuit designs for the Quantum Query and Key Projections are presented in figure 4, while the Quantum Value Projection is detailed in figure 5. After a measurement step, the resulting classical latent representations (Q_{latent}, K_{latent}, V_{latent}) are passed to the Gaussian Projected Quantum Multi-head Attention (GPMHA) mechanism. The output is then processed through a residual connection and layer normalization (Add & Norm), followed by a Quantum Feed-Forward Network (QFFN). A second Add & Norm layer is applied before a final linear transformation and a softmax function produce the output probabilities.

achieved competitive results with the FNet model. This result validates the potential of the novel quantum architecture, although its performance does not yet surpass that of the standard Transformer model.

3. FitQT: the proposed model

In this section, we provide a detailed description of our proposed model, FitQT, short for the Fully Integrated and Trainable Quantum Transformer. The name encapsulates the core design principles of our work: to develop a structurally complete, robustly trainable, and scalable architecture suitable for complex Natural Language Processing tasks. Our approach is distinguished by the synergistic integration of three pivotal quantum components that form a cohesive whole. The architecture features a novel Gaussian Projected Quantum Multi-head Attention (GPMHA) mechanism to capture diverse contextual relationships, a Quantum Feed-Forward Network (QFFN) to enhance non-linear expressivity, and, critically, quantum residual connections to ensure stable gradient flow and facilitate the training of deeper circuits. This holistic design marks a step beyond fragmented component-level studies, creating a powerful and practical framework for the QNLP domain. The overall architecture of the FitQT is presented in figure 3. The principal components of FitQT are discussed in the following subsections.

3.1. Gaussian projected quantum multi-head attention architecture

Our primary contribution is the development of a novel, native quantum attention mechanism, termed Gaussian Projected Quantum Multi-head Attention Architecture (GPMHA). This mechanism fundamentally

re-architects the principles of self-attention for quantum computing models. The architecture is meticulously designed to leverage the expressive power of high-dimensional Hilbert space while circumventing the inherent challenges of direct state vector manipulation and fidelity estimation in the Noisy Intermediate-Scale Quantum (NISQ) era. The entire mechanism is encapsulated within a modular Quantum Self-Attention Layer.

3.1.1. Quantum state preparation and feature encoding

The initial stage of our proposed process involves transforming the classical input data, represented as a sequence of vectors $y_s^{(l-1)} \in \mathbb{R}^d$ at layer $(l-1)$, into quantum states. This is executed via a hybrid quantum–classical framework in which a parameterized quantum circuit is integrated within a classical neural network layer.

Specifically, the quantum circuit is designed to accept two distinct types of arguments: the classical feature vectors and a set of trainable variational parameters. Inside the circuit, the input data $y_s^{(l-1)}$ is used to parameterize a sequence of quantum gates in a block known as the 'encoding circuit,' U_{enc}. This technique, also referred to as 'angle encoding' or 'data re-uploading,' acts as a nonlinear feature map. Subsequently, the trainable parameters θ are used to drive a 'variational circuit' block, U_{var}. The final n-qubit quantum state, $|\psi_s\rangle$, is therefore prepared by sequentially applying both of these circuit blocks to an initial reference state (e.g., the $|0^n\rangle$ state passed through Hadamard gates). This process can be generally represented as:

$$|\psi_s\rangle = U_{\text{var}}(\theta)\, U_{\text{enc}}(y_s^{(l-1)})\, H^{\otimes n}|0^n\rangle, \tag{9}$$

Here, the preparation of the final quantum state $|\psi_s\rangle$ begins with an initial n-qubit reference state $|0^n\rangle$, where all qubits are in the ground state. A layer of Hadamard gates $H^{\otimes n}$ is first applied to create a uniform superposition, preparing the register to effectively embed information across its state space. Subsequently, the data-dependent encoding unitary $U_{\text{enc}}(y_s^{(l-1)})$ imprints the classical feature vector $y_s^{(l-1)}$ onto the quantum state. Finally, the parameter-dependent variational unitary $U_{\text{var}}(\theta)$, driven by the set of trainable parameters θ, transforms the encoded state in a learnable manner. This clear separation between the data-dependent U_{enc} and the parameter-dependent U_{var} allows classical deep learning frameworks to effectively track and optimize the parameters θ, while still leveraging the input data to drive the quantum transformation. This structured approach maps the classical data into a rich quantum feature space, setting the stage for the subsequent quantum processing steps.

3.1.2. Asymmetric latent projections

A central tenet of FitQT is the asymmetric projection of the encoded quantum state into classical latent representations for the Query, Key, and Value components. This approach diverges significantly from the homogeneous linear transformations in classical attention mechanisms. We employ three distinct variational circuits, defined by parameter sets θ_q, θ_k, and θ_v, to generate these representations.

To foster stable and efficient training and to mitigate the effects of barren plateaus, the variational parameters for these transformations are initialized within a constrained domain, typically sampled from a normal distribution and scaled to a small interval around zero.

The Query and Key representations are derived via a projection that collapses the quantum state's relational information onto a low-dimensional manifold. For an input state $|\psi_s\rangle$, the latent scalar Query and Key are computed as expectation values of a single, fixed observable, here chosen as the Pauli-Z operator acting on the first qubit Z_1:

$$Q_{\text{latent}}(s) \coloneqq \langle\psi_s|U_{\text{proc}}(\theta_q)^\dagger Z_1 U_{\text{proc}}(\theta_q)|\psi_s\rangle \tag{10}$$

$$K_{\text{latent}}(s) \coloneqq \langle\psi_s|U_{\text{proc}}(\theta_k)^\dagger Z_1 U_{\text{proc}}(\theta_k)|\psi_s\rangle \tag{11}$$

In these equations, a trainable quantum circuit, denoted as the processing unitary U_{proc}, is applied to the input state $|\psi_s\rangle$. This circuit is parameterized by an independent set of variables for Query (θ_q) and Key (θ_k). The term $U_{\text{proc}}^\dagger$ represents the conjugate transpose of the unitary, and the entire expression $\langle\psi_s|\ldots|\psi_s\rangle$ calculates the expectation value of the Pauli-Z observable (Z_1) after the state has been transformed. By measuring only a single, fixed observable on the first qubit, this design compels the model to learn quantum transformations that distill the relational essence of an input into a single scalar value.

The Value representation, in contrast, is designed to preserve the rich semantic content of the quantum state. It is constructed as a high-dimensional vector whose elements are the expectation values of a set of predefined, non-trainable Pauli-Z observables $\{P_j\}$:

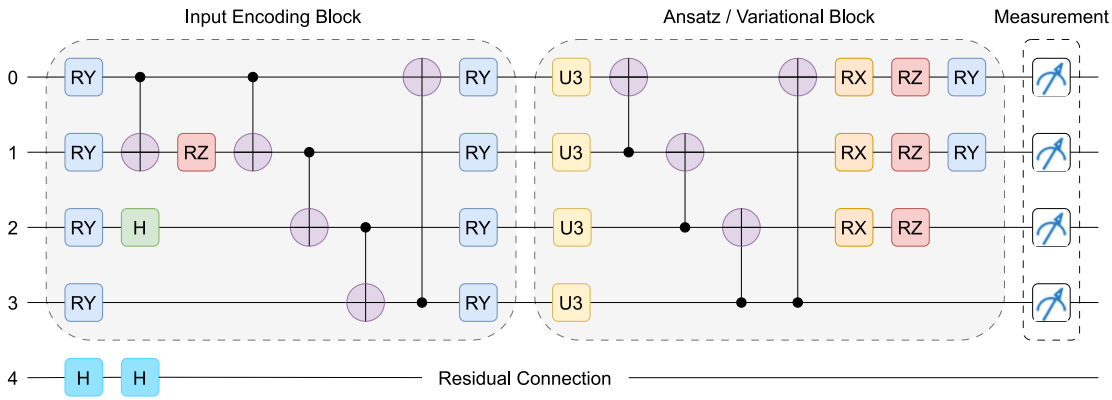


Figure 4. Detailed schematic of the quantum circuit architecture for generating Query (Q) and Key (K) features. This circuit is divided into three main blocks: (1) the Input Encoding Block, which encodes classical input data into quantum states; (2) the Ansatz/Variational Block, which applies a sequence of parameterized unitary transformations, including a Residual Connection mechanism utilizing an ancillary qubit (ancilla qubit, wire 4) to enhance trainability; and (3) the Measurement Block, which performs measurements on the main qubits. Notably, this same variational quantum circuit architecture is used to generate the latent features for both Query and Key, differing only in the set of trainable parameters employed (θ_q for Query and θ_k for Key).

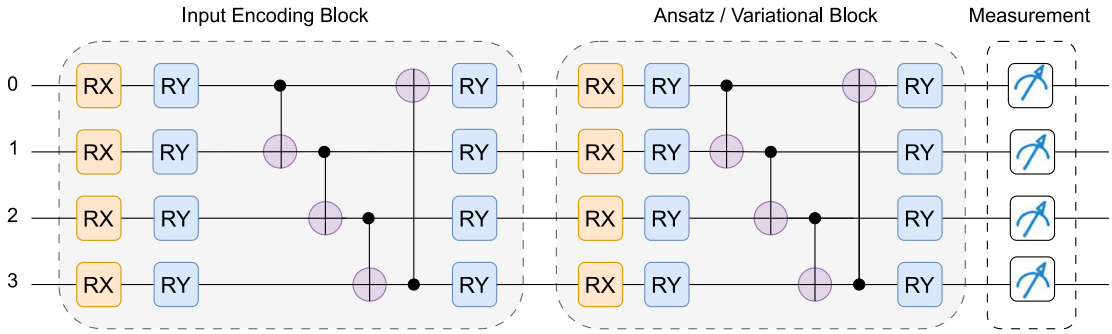


Figure 5. Quantum circuit architecture for generating the Value (V) feature in the GPMHA mechanism. This circuit processes an encoded quantum state (similar to Q/K generation) through a distinct Ansatz/Variational Block (Uvar()), comprising parameterized rotation (RX, RY) and entangling (CNOT) layers. Unlike Q and K, the Value feature is designed to preserve rich semantic information. Consequently, the Measurement Block performs expectation value measurements of a diverse set of pre-defined, non-trainable Pauli string observables on the main qubits, yielding a multi-dimensional Value feature vector. This architecture ensures the Value carries the necessary contextual information payload for the attention mechanism.

$$V_{\text{latent}}(s) := \begin{bmatrix} \langle P_1 \rangle_s \\ \langle P_2 \rangle_s \\ \vdots \\ \langle P_{d_v} \rangle_s \end{bmatrix} \tag{12}$$

where each element $\langle \psi_s | U_{\text{proc}}(\theta_v)^\dagger P_j U_{\text{proc}}(\theta_v) | \psi_s \rangle$ is calculated. Here, another distinct processing unitary, $U_{\text{proc}}(\theta_v)$, is learned specifically for the Value component. Instead of a single observable, a set of d_v different Pauli string observables, $\{P_j\}$, is measured. This allows for the extraction of a much richer, multi-faceted representation from the quantum state, forming a high-dimensional classical vector.

This design philosophy establishes a clear functional separation: Query and Key vectors serve as low-dimensional 'pointers' for determining relevance, while the Value vector acts as a high-dimensional 'payload' of contextual information. The circuit structures for generating the quantum Query, Key, and Value representations are illustrated in figures 4 and 5.

3.1.3. Hybrid multi-head attention mechanism

FitQT utilizes a multi-head attention (MHA) mechanism to capture complex relational patterns inside a resource-efficient hybrid quantum–classical framework. The quantum circuit is performed once for each input token to produce a unified collection of latent Query, Key, and Value representations. These are subsequently transmitted to various classical 'attention heads'.

Within each head h, these shared latent representations are first passed through head-specific linear transformations to produce the final Query, Key, and Value vectors for that head. The attention scores, which

quantify inter-token relevance, are computed not via the conventional dot product but through a scaled Gaussian kernel:

$$\alpha_{s,j,h} \coloneqq \exp\left(-\frac{\|Q_h(s) - K_h(j)\|^2}{\tau}\right) \tag{13}$$

where τ is a temperature scaling parameter, typically related to the dimension of the key vectors. Following normalization, these scores produce the attention weights, which facilitate the calculation of a weighted sum of the Value vectors, culminating in an output head_h for each head. The outputs from all heads are subsequently concatenated and processed by a final output projection matrix, W_o, to yield a consolidated attention output, $O_{\text{attention}}$:

$$O_{\text{attention}} = \text{Concat}(\text{head}_1, \ldots, \text{head}_H) \cdot W_o \tag{14}$$

This unified output is subsequently integrated with the original input vector through a residual connection, a standard technique for stabilizing deep network training:

$$y_s^{(l)} = y_s^{(l-1)} + O_{\text{attention}}(s) \tag{15}$$

This hybrid architecture enhances expressive power while confining the quantum computational cost to a single, shared feature extraction stage, rendering the model scalable and suitable for NISQ-era devices.

3.2. Quantum residual connection

Within the proposed quantum model architecture, a quantum residual connection mechanism is intricately integrated into the parameterized ansatz layers to enhance trainability and expressive power. The variational circuit, denoted as $U_{\text{var}}(\boldsymbol{\theta})$, is constructed from D layers. Each layer typically consists of a sub-layer of entangling gates, such as CNOTs arranged in a cyclic pattern, followed by a sub-layer of parameterized single-qubit rotations. To ensure high expressivity, these single-qubit rotations are often implemented using general U_3 gates, $U_3(\theta_{\text{rot}}, \phi_{\text{rot}}, \lambda_{\text{rot}})$, which can realize any arbitrary single-qubit unitary transformation by adjusting its three Euler angle parameters. These angles form a subset of the trainable parameters $\boldsymbol{\theta}$. This mechanism is realized through the Linear Combination of Unitaries (LCU) technique, leveraging a single ancillary qubit. Specifically, at a designated layer within the ansatz block—which comprises a sequence of parameterized single- and multi-qubit quantum gates, collectively denoted as $W(\theta)$—the quantum residual connection is implemented as follows.

Initially, the ancillary qubit, prepared in the ground state $|0\rangle_a$ (where a denotes the ancilla), is transformed into an equal superposition state by the application of a Hadamard gate H_a:

$$H_a|0\rangle_a = \frac{|0\rangle_a + |1\rangle_a}{\sqrt{2}} \tag{16}$$

Subsequently, a unitary block $U_{\text{block}}(\theta)$, representing a segment of the overall ansatz $W(\theta)$ and embodying the evolution governed by the trainable parameters θ, is applied to the main system qubits. The application of $U_{\text{block}}(\theta)$ is conditioned on the state of the ancillary qubit, forming a controlled unitary operation $C\text{-}U_{\text{block}}$. This can be expressed as:

$$C\text{-}U_{\text{block}} = |0\rangle_a\langle 0|_a \otimes I_{\text{main}} + |1\rangle_a\langle 1|_a \otimes U_{\text{block}}(\theta)_{\text{main}} \tag{17}$$

where $|0\rangle_a\langle 0|_a$ and $|1\rangle_a\langle 1|_a$ are projection operators onto the ancilla's basis states, $\otimes$ denotes the tensor product, and I_{main} is the identity operator on the Hilbert space of the main (computational) qubits.

To complete the LCU framework and facilitate quantum interference, a second Hadamard gate H_a is applied to the ancillary qubit. If $|\psi_{\text{in}}\rangle$ represents the state of the main qubits prior to this residual block, the state of the composite system (main and ancillary qubits) evolves to:

$$\frac{1}{2}[(I_{\text{main}} + U_{\text{block}}(\theta))|\psi_{\text{in}}\rangle\ |0\rangle_a + (I_{\text{main}} - U_{\text{block}}(\theta))|\psi_{\text{in}}\rangle\ |1\rangle_a] \tag{18}$$

Upon projective measurement of the ancillary qubit and post-selection for the outcome $m \cdot a = 0$ (corresponding to the ancilla being measured in the state $|0\rangle_a$, the effective operator $R(\theta)$ acting on the input state $|\psi_{\text{in}}\rangle$ of the main qubits is given by:

$$R(\theta)|\psi_{\text{in}}\rangle = \frac{1}{2}(I_{\text{main}} + U_{\text{block}}(\theta))|\psi_{\text{in}}\rangle \tag{19}$$

This expression clearly illustrates the residual structure: the input state $|\psi_{\text{in}}\rangle$ (via the term $\frac{1}{2}I_{\text{main}}|\psi_{\text{in}}\rangle$) is superposed with its transformed counterpart $\frac{1}{2}U_{\text{block}}(\theta)|\psi_{\text{in}}\rangle$. The introduction of the ancillary qubit and the LCU construction effectively establishes a parallel quantum channel, allowing an 'unevolved' component of the

system's state to be preserved and coherently combined with the component that has undergone the parameterized quantum evolution.

A significant advantage of this design is the inherent capability of the evolutionary block $U_{\text{block}}(\theta)$ to learn to approximate the identity operator I_{main} (i.e., $U_{\text{block}}(\theta) \approx I_{\text{main}}$) during the optimization process. In such a scenario, the effective residual operator $R(\theta)$ naturally converges towards $R(\theta) \approx I_{\text{main}}$, enabling the quantum signal to propagate through the layer with minimal alteration. This characteristic mirrors the ability of residual layers in classical neural networks to learn identity mappings, which is crucial for mitigating training challenges such as vanishing gradients and barren plateaus. Consequently, this facilitates the construction and training of deeper and more complex quantum circuits, thereby unlocking the potential for quantum models with superior expressive capabilities.

3.3. Quantum feed-forward network

In the domain of quantum machine learning, the development of neural network architectures possessing robust expressive capabilities is a pivotal objective. Drawing inspiration from the successes of classical deep neural networks, FitQT proposes a fundamental quantum processing unit, termed a Quantum Feed-Forward Network (QFFN) block, designed to emulate the functionality of feed-forward layers found in classical architectures. A significant feature of the proposed QFFN block is the integration of a non-linear activation mechanism, engineered to mimic the behavior of the Gaussian Error Linear Unit (GELU) function—a popular activation in modern deep learning models due to its smoothing properties and superior performance. The entire QFFN block is conceptualized to operate within a predominantly quantum framework, encompassing stages for input data encoding, parameterized quantum transformations, a distinct quantum non-linear activation phase, and finally, a measurement process for information extraction.

The operational flow of the QFFN block commences with the conversion of classical input data, typically an N-dimensional feature vector $x \in \mathbb{R}^N$, into a quantum state within an N-qubit system. This stage, known as quantum encoding, is essential for embedding classical information into the high-dimensional Hilbert space of the quantum system. A prevalent encoding strategy is 'angle encoding,' where each component x_i of the input vector is used to determine the rotation angle of one or more single-qubit quantum gates acting on the ith qubit or a subset of qubits. For example, if the system is initialized in the ground state $|0\rangle^{\otimes N}$, the encoding unitary $U_{\text{enc}}(x)$ can be expressed as a tensor product of single-qubit rotations, such as

$$U_{\text{enc}}(x) = \bigotimes_{i=1}^{N} R_y(f(x_i))_i, \tag{20}$$

where $R_y(\phi)_i$ is a rotation about the Y-axis of the ith qubit by an angle ϕ, and $f(x_i)$ is a (possibly identity) preprocessing function of the data component x_i. The result of this stage is a quantum state

$$|\psi_{\text{enc}}\rangle = U_{\text{enc}}(x)\,|0\rangle^{\otimes N}, \tag{21}$$

which carries the information of x within its complex amplitudes.

Following data encoding, the quantum state $|\psi_{\text{enc}}\rangle$ undergoes a first parameterized quantum transformation layer, described by a unitary operator $U^{(1)}_{\text{trans}}(\alpha_1)$. This layer serves a role analogous to a linear layer in classical networks, but the transformations are effected by quantum gates. The structure of $U^{(1)}_{\text{trans}}(\alpha_1)$ typically includes a sequence of single-qubit rotation gates, whose rotation angles are trainable parameters $\alpha_{1,k}$, and multi-qubit gates to induce entanglement among the qubits. Such parameterized ansatz architectures are often referred to as Variational Quantum Circuits (VQCs). For example, $U^{(1)}_{\text{trans}}(\alpha_1)$ might be constructed by repeating a basic block of R_x, R_y, R_z gates on each qubit, followed by a layer of CNOT or CZ gates arranged in a specific topology (e.g., linear, circular, or all-to-all) to ensure effective inter-qubit interaction. The parameters α_1 within $U^{(1)}_{\text{trans}}$ are variables optimized during the model's training process. The state after this transformation layer is

$$|\psi^{(1)}_{\text{trans}}\rangle = U^{(1)}_{\text{trans}}(\alpha_1)\,|\psi_{\text{enc}}\rangle. \tag{22}$$

The subsequent, and arguably most challenging, stage in constructing quantum neural networks is the quantum activation phase, represented by the unitary operator U_{act}. The objective of this stage is to introduce a non-linear element into the processing, similar to the role of activation functions in classical neural networks. Non-linearity is key to the powerful representational capacity of classical deep learning models, enabling them to learn complex relationships in data. However, quantum evolutions, described by unitary operators, are inherently linear. Therefore, realizing an effective non-linear transformation within a fully unitary quantum framework requires sophisticated approaches. In this research, we focus on simulating the GELU activation function.

One strategy for realizing U_{act} involves approximating the GELU function with a polynomial

$$P_{\text{GELU}}(z) = \sum_{k=0}^{d_p} c_k z^k \tag{23}$$

of degree d_p. Once such a polynomial approximation is obtained, advanced techniques like Quantum Signal Processing (QSP) or its generalization, Quantum Singular Value Transformation (QSVT), can be employed.

QSP/QSVT offers a robust framework for implementing polynomial transformations on the singular values (or eigenvalues in certain instances) of a unitary operator or a block encoded within an expanded unitary. This process typically involves 'block-encoding' an operator A_i (related to the 'activation' of the ith qubit prior to the non-linear stage) into a larger unitary matrix U_A acting on both the main quantum system and some ancillary qubits. Subsequently, a sequence of single-qubit rotations (signal processing unitaries) is applied to an ancillary qubit, interleaved with conditional applications of U_A and $U_A^\dagger$ (its inverse). The angles of these signal processing rotations are carefully determined from the coefficients c_k of the polynomial P_{GELU}. The resulting operator

$$U_{\text{act}} \approx U(P_{\text{GELU}}(A)) \tag{24}$$

then coherently implements the desired polynomial transformation on the quantum 'signals.' While theoretically powerful, the implementation of QSP/QSVT demands significant quantum resources. An alternative, simpler approach that breaks full quantum coherence involves intermediate measurements and classical feedback. The quantum state after the activation stage is denoted

$$|\psi_{\text{act}}\rangle = U_{\text{act}}|\psi^{(1)}_{\text{trans}}\rangle. \tag{25}$$

Following the non-linear activation stage, a second parameterized quantum transformation layer, characterized by the unitary operator $U^{(2)}_{\text{trans}}(\alpha_2)$, is applied to the state $|\psi_{\text{act}}\rangle$, yielding the state

$$|\psi_{\text{final}}\rangle = U^{(2)}_{\text{trans}}(\alpha_2)|\psi_{\text{act}}\rangle. \tag{26}$$

This layer shares a similar structure to the first transformation layer but possesses an independent set of trainable parameters α_2. The presence of multiple parameterized transformation layers interspersed with non-linear activation mechanisms allows the model to learn increasingly abstract and complex data representations, analogous to classical deep neural networks.

Ultimately, a measurement process is conducted to recover classical information from the final quantum state $|\psi_{\text{final}}\rangle$. Typically, this involves computing the expectation values of a set of observables, $\{M_k\}$, with respect to the final state. These observables can be single-qubit Pauli operators (e.g., $\sigma_z^{(i)}$ on the ith qubit) or more complex Pauli strings. The output of the QFFN block is a classical vector y, with each component, y_k, being the expectation value of a corresponding observable M_k with respect to the final state, i.e.,

$$y_k = \langle\psi_{\text{final}}|M_k|\psi_{\text{final}}\rangle.$$

This vector $\mathbf{y}$ then serves as the output of the network block, which can be used for subsequent classical post-processing. For example, in a classification task, this output vector might be passed through a softmax function to yield class probabilities, or it could be used to compute a loss function for optimizing the trainable parameters α_1 and α_2 during the learning process. The detailed training algorithm is presented in algorithm 1.

Algorithm 1. FitQT training algorithm.

Require:
- Training dataset $\mathcal{D} := \{(\mathbf{x}^{(m)}, y^{(m)})\}_{m=1}^{N_s}$, where:
  - $\mathbf{x}^{(m)} \in \mathbb{R}^n$: input feature vector
  - $y^{(m)} \in \{0, 1\}$: binary label
- Number of training epochs: N_{epochs}
- QSAL parameters: L_{QSAL}, D_{enc}, D_{var}, N_{heads}
- QFF (Quantum Feed Forward) parameters: D_{QFF}
- Optimizer: AdamW with learning rate η, weight decay λ
- Loss: Binary Cross-Entropy

Ensure:
- Trained model parameters: Θ^*, W^*, B^*
- Model $f_{\Theta^*,W^*,B^*}(\mathbf{x})$ for predictions $\hat{y} \in [0, 1]$

1: **procedure** TrainFitQT
2: Initialize model parameters Θ, W, B
3: Initialize AdamW optimizer and CrossEntropyCriterion
4: **for** epoch = 1 to N_{epochs} **do**
5: **for** sample $m = 1$ to N_s **do**
6: $\mathbf{h}_s^{(0)} \leftarrow$ InitialFeatureEncoder$(\mathbf{x}_s^{(m)})$, for $s = 1..S$

(Continued.)

for $l = 1$ to L_{QSAL} **do**
  // Step 1: Shared Quantum Feature Extraction
  for $s = 1$ to S **do**
    Encode classical vector $\mathbf{h}_s^{(l-1)}$ into a quantum state $|\psi_s^{(l-1)}\rangle$
    Apply residual quantum blocks $U_q^{(l)}$, $U_k^{(l)}$, $U_v^{(l)}$ to the state $|\psi_s^{(l-1)}\rangle$ and measure to obtain latent features: $Q_{latent}^{(l)}(s)$, $K_{latent}^{(l)}(s)$, $V_{latent}^{(l)}(s)$
  // Step 2: Gaussian Multi-Head Attention
  for $s = 1$ to S **do**
    Initialize list of head outputs: head_outputs $\leftarrow$ []
    for $h = 1$ to N_{heads} **do**
    Project latent features for head h: Q_h, K_h, $V_h \leftarrow$ Project$_h(Q_{latent}^{(l)}, K_{latent}^{(l)}, V_{latent}^{(l)})$
    Compute weights α_h using Gaussian Kernel on (Q_h, K_h)
    Compute head output head$_h$ as weighted sum of V_h using normalized weights α_h
    Append head$_h$ to head_outputs
    Concatenate and project all head outputs: $o_s^{(l)} \leftarrow$ Concat(head_outputs) $\cdot W_O$
    Add classical residual connection: $\mathbf{h}_s^{(l)} \leftarrow \mathbf{h}_s^{(l-1)} + o_s^{(l)}$
  // Step 3: Quantum Feed-Forward Layer
  for $s = 1$ to S **do**
    $\mathbf{z}_s^{(0)} \leftarrow \mathbf{h}_s^{(L_{QSAL})}$
    Apply residual Quantum Feed-Forward block (QFF) on $\mathbf{z}_s^{(0)}$ to get $\mathbf{z}_s^{(QFF)}$
    Add **classical** residual connection: $\mathbf{z}_s^{(QFF-final)} \leftarrow \mathbf{z}_s^{(0)} + \mathbf{z}_s^{(QFF)}$
  // Step 4: Aggregation, Prediction, and Update
  Aggregate token representations: $\mathbf{h}_{final}^{(m)} \leftarrow$ AveragePooling$(\{\mathbf{z}_s^{(QFF-final)}\})$
  Generate prediction: $\hat{y}^{(m)} \leftarrow$ FullyConnectedLayer$(\mathbf{h}_{final}^{(m)})$
  Compute loss: $\mathcal{L}^{(m)} \leftarrow$ CrossEntropyCriterion$(\hat{y}^{(m)}, y^{(m)})$
  Update parameters Θ, W, B using optimizer and $\nabla\mathcal{L}^{(m)}$
  if stopping condition met **then break**
return Trained parameters Θ^*, W^*, B^*

4. Experimental setup and results

4.1. Experimental environment

All experiments were conducted using the Kaggle cloud-based platform, which provides access to high-performance computing resources. Specifically, a virtual machine equipped with an NVIDIA Tesla P100 GPU, Intel Xeon CPU, and 16 GB of RAM was utilized for training and evaluation. The environment was configured with internet access enabled to support the loading of pre-trained language models and relevant datasets. All code executions were performed within a Kaggle notebook session with GPU acceleration.

4.2. Evaluation metrics

To assess model performance, four standard metrics are employed: Accuracy, Precision, Recall, and F1-score. These metrics provide a comprehensive evaluation, especially in scenarios with imbalanced class distributions. Where applicable, weighted averaging is used to account for class imbalance, ensuring each class contributes proportionally based on its frequency in the dataset.

Accuracy indicates the proportion of correctly classified instances among all predictions. It is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where

- TP is the number of true positives,
- TN is the number of true negatives,
- FP is the number of false positives, and
- FN is the number of false negatives.

Precision quantifies the proportion of true positive predictions among all positive predictions made by the model. It is particularly important when the cost of false positives is high. The formula is:

Table 1. Classical architecture and training hyperparameters (shared across tasks).

Hyperparameter	Value	Description
Model Architecture		
Number of Layers (`num_layers`)	2	Number of stacked Quantum Self-Attention Layers (QSAL).
Number of Attention Heads (`num_heads`)	32	Number of classical Gaussian attention heads.
Training Configuration		
Learning Rate	1×10^{-2}	Step size for updating classical parameters.
Weight Decay	1×10^{-4}	L2 regularization coefficient.
Optimizer	AdamW	Classical optimizer used for training.
Loss Function	CrossEntropy	Supervised classification loss.
Number of Epochs	30 / 60	English SA / Vietnamese SA, respectively.
Early Stopping Patience	5	Training is terminated if the training loss does not improve for 5 consecutive epochs.

Table 2. Quantum resource requirements per QSAL layer in FitQT (shared across tasks).

Quantum resource metric	Value	Description
Qubit Configuration (per layer)		
Computational Qubits (n)	4	Number of data-carrying qubits per QSAL layer.
Ancilla Qubits	1	Single ancillary qubit used for LCU-based residual connection (reused).
Total Qubits	5	Fixed qubit count, independent of model depth.
Circuit Complexity (per layer)		
QSAL Circuit Depth	33	Depth of Q/K/V projection and measurement circuit.
Residual Connection Overhead	O(1)	Constant-depth overhead introduced by LCU residual connection.
QFFN Circuit Depth	O(1)	Fixed-depth variational block used in QFFN.
Measurement Overhead (per forward pass)		
Measurements per Sample	22	One measurement for Q, one for K, and multiple Pauli observables for V.
Shots per Measurement	1,000	Number of shots to achieve statistical error $\epsilon \approx 0.01$.
Total Shots (batch = 32)	704,000	Hardware-level estimate per forward pass.
Scalability and Feasibility		
Ancilla Scaling with Depth	O(1)	Ancilla usage does not increase with the number of layers.
NISQ Compatibility	Yes	Compatible with current NISQ platforms.

$$\text{Precision} = \frac{TP}{TP + FP}$$

In multi-class settings, weighted precision is computed by averaging the precision of each class, weighted by the number of true instances per class.

Recall, also known as sensitivity, measures the proportion of actual positive instances that were correctly identified by the model. It is especially critical in applications where missing a positive case is costly. The recall is calculated as:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Weighted recall considers the relative size of each class to provide a balanced evaluation.

F1-score is the harmonic mean of precision and recall, offering a single metric that balances both. It is useful when there is a trade-off between precision and recall, particularly in imbalanced datasets. The F1-score is defined as:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

In multi-class classification, the weighted F1-score aggregates the individual scores from each class based on their respective support (number of true instances).

4.3. Hyperparameters

The hyperparameters and configuration settings of the proposed FitQT model are summarized in tables 1 and 2. Specifically, table 1 reports the classical architecture and training hyperparameters used for the English and Vietnamese sentiment analysis tasks, including the number of attention heads and optimization settings. In contrast, table 2 presents a detailed analysis of the quantum circuit configuration and resource requirements

Table 3. Dataset statistics for English and Vietnamese sentiment analysis.

Dataset	Language	Domain	Total samples	Train/Val/Test Split
Yelp	English	Restaurant review	1,000	8/0/2
IMDb	English	Movie review	1,000	8/0/2
Amazon	English	Product review	1,000	8/0/2
UIT-VSFC	Vietnamese	Student feedback	16,175	7/1/2

Table 4. Comparison of test accuracy and number of parameters for Vanilla Transformer [27], QSANN [43], QMSAN [69] and FitQT on Yelp, IMDb, and Amazon datasets. The results for the Vanilla Transformer are reported by the QSANN's authors. This comparison focuses on models that incorporate quantum-inspired or quantum-based architectures.

Method	Yelp		IMDb		Amazon	
	# Params	Accuracy (%)	# Params	Accuracy (%)	# Params	Accuracy (%)
CSANN [27]	785	83.11 ± 0.89	785	79.67 ± 0.83	785	83.22 ± 1.28
QSANN [43]	49	84.79 ± 1.29	49	80.28 ± 1.78	61	84.25 ± 1.75
QMSAN [69]	71	84.96 ± 3.34	71	84.29 ± 2.32	71	87.48 ± 1.02
FitQT (ours)	2595	**92.94 ± 0.16**	2595	**95.17 ± 0.71**	2595	**89.72 ± 0.25**

per Quantum Self-Attention Layer (QSAL), covering qubit counts, circuit depth, measurement overhead, and NISQ feasibility.

4.4. Datasets and preprocessing

We use both English and Vietnamese sentiment analysis datasets to evaluate FitQT in comparison with other quantum- and non-quantum-based models. For English sentiment analysis, three available datasets introduced by [71] were used. These datasets comprise user reviews for restaurants (Yelp), movies (IMDb), and products (Amazon). Each dataset contains 1000 textual sequences, evenly balanced with 500 sentences labeled as negative ('0') and 500 as positive ('1'). Sequence lengths vary from a few words to several dozen. For each dataset, 80% of the sequences were randomly allocated for training and the remaining 20% for testing. This experimental setup is analogous to that used in the study [43], facilitating a direct performance comparison of the proposed quantum transformer method against their findings. For Vietnamese sentiment analysis, we use the UIT-VSFC dataset [72], compiled from student feedback collected at a Vietnamese university between 2013 and 2016. The data was gathered through an automated online survey system administered at the end of each academic semester. These surveys combined closed-ended questions using a 5-point Likert scale with open-ended questions that allowed students to provide more detailed comments. In total, the dataset contains 16,175 feedback sentences, each annotated with two types of labels: sentiment and topic. Sentiment labels are categorized into three classes: negative (0), neutral (1), and positive (2). Topic labels are divided into four groups: Lecturer (0), Curriculum (1), Facility (2), and Others (3). Table 3 summarizes these datasets.

To ensure fair comparison and reproducibility across languages and models, a unified input preprocessing and feature encoding pipeline is adopted for all datasets. For English datasets, the pretrained BERT-base-uncased model is used as a sentence-level feature encoder, while for Vietnamese datasets, PhoBERT is employed in place of BERT. Apart from this substitution of the pretrained language model, all preprocessing steps remain identical across languages. Text inputs are tokenized using the corresponding tokenizer, truncated to a maximum sequence length of 256 tokens, and dynamically padded within each batch. Sentence representations are extracted using the [CLS] token embedding from the final hidden layer. In all experiments, the pretrained language models are used solely as frozen feature encoders and are not fine-tuned during training. The resulting fixed-dimensional embeddings serve as classical input features for the proposed FitQT model as well as for all comparative baseline methods.

4.5. Comparison with quantum-inspired baselines on the Yelp, IMDb, and amazon datasets

The experiments were conducted under experimental settings consistent with those described in the corresponding baseline papers to ensure fair comparison. For models where official implementations or reported results were available, we adopted the performance values reported in the original publications. Table 4 summarizes the comparison between FitQT and existing quantum-inspired architectures.

Table 5. Test accuracy (%) comparison across datasets.

Model	Yelp Acc. (%)	IMDb Acc. (%)	Amazon Acc. (%)
BERT Tiny [73]	84.61 ± 1.31	78.67 ± 1.76	82.17 ± 1.11
BERT Mini [73]	87.94 ± 1.61	84.83 ± 1.03	83.89 ± 1.35
BERT Small [73]	91.67 ± 0.94	85.50 ± 0.75	84.50 ± 1.11
BERT Medium [73]	91.94 ± 1.26	89.89 ± 0.84	87.50 ± 1.25
BERT Base [74]	**95.44 ± 0.68**	95.11 ± 0.77	**90.50 ± 0.75**
DistilBERT [75]	94.44 ± 0.44	94.39 ± 0.52	90.11 ± 0.46
FitQT (ours)	92.94 ± 0.16	**95.17 ± 0.71**	89.72 ± 0.25

Table 6. Training efficiency comparison across Yelp, IMDb, and Amazon datasets. All values are averaged over multiple independent runs.

Model	#Params (M)	Yelp			IMDb			Amazon		
	Ep.	Total (s)	/Ep.	Ep.	Total (s)	/Ep.	Ep.	Total (s)	/Ep.	
BERT Tiny	4.4	60	33	0.52	60	34.8	0.55	60	33	0.52
BERT Mini	11.2	28	43.8	1.50	49	76.2	1.50	18	54.6	1.50
BERT Small	28.8	21	73.8	3.37	18	64.2	3.38	16	56.4	3.35
BERT Medium	41.4	16	108	6.54	13	87.6	6.54	12	80.4	6.52
BERT Base	108.3	11	225	19.96	11	225.6	19.98	12	246	20
DistilBERT	66.4	12	125.4	10.11	14	144.6	10.04	12	124.8	10.10
FitQT (ours)	**0.0026**	60	61.2	1.02	60	72	1.2	60	66.6	1.11

As shown in the table, FitQT achieves the highest accuracy across all three datasets while maintaining relatively low standard deviation values. The proposed model contains 2,595 classical parameters, which is larger than QSANN and QMSAN due to architectural extensions introduced in the hybrid design.

4.6. Comparison with transformer architectures on the Yelp, IMDb, and amazon datasets

In addition to comparisons with quantum-inspired baselines such as CSANN and QSANN, we evaluated FitQT against standard Transformer architectures and their compact variants, including BERT Tiny, Mini, Small, Medium, DistilBERT, and BERT Base. It is worth noting that pretrained models like BERT Base benefit from large-scale self-supervised pretraining prior to task-specific fine-tuning, whereas FitQT is trained from scratch using only task-labeled data. Therefore, the observed competitive performance should be interpreted in light of these differences in training paradigms rather than as a direct indication of architectural equivalence.

Regarding model size, FitQT contains 2,595 classical parameters. However, the architecture incorporates four quantum-inspired qubits as part of its hybrid design. While these components enhance representational capacity, they introduce computational mechanisms that differ from conventional parameter-based neural networks. Consequently, parameter count alone may not fully capture the computational characteristics of the hybrid quantum–classical framework.

Table 5 presents the evaluation results of several transformer-based models across three benchmark sentiment analysis datasets: Yelp, IMDb, and Amazon. Our proposed model, FitQT, achieves test accuracies of 92.94%, 95.17%, and 89.72% on Yelp, IMDb, and Amazon respectively, which are highly competitive compared to larger models such as BERT-Medium and DistilBERT. While BERT-Medium requires over 41 million parameters and DistilBERT over 66 million, FitQT attains comparable accuracy with only 2,595 parameters.

Importantly, all models were trained and evaluated under the same experimental setup, ensuring that the reported results were obtained under consistent conditions rather than sourced from external studies.

To further analyze computational efficiency, table 6 reports the number of parameters, training epochs, total training time, and average time per epoch, averaged over multiple independent runs.

From the results, FitQT requires approximately 1.02–1.20 s per epoch, which is substantially lower than BERT-Medium ($\approx$ 6.53 s) and DistilBERT ($\approx$ 10.08 s), though higher than BERT Tiny. Despite its compact size, the model achieves task-level accuracy close to that of BERT-Base on IMDb (95.17% versus 95.11%), while using less than 0.01% of the number of parameters.

Overall, the results indicate that the proposed hybrid quantum-inspired architecture achieves a favorable balance between predictive performance and computational cost. The combination of competitive accuracy and reduced parameter size suggests its potential applicability in scenarios where model compactness and training efficiency are important considerations.

Table 7. Ablation study of individual architectural components on the IMDb dataset.

Model variant	Total Parameters	Accuracy
Quantum Single-Head Attention	17	76.61 ± 0.53
Quantum Single-Head + QFFN	18	81.94 ± 0.61
Quantum Multi-Head Attention	2,529	86.22 ± 1.00
Quantum Multi-Head + QFFN	2,533	92.83 ± 0.56
Quantum Multi-Head + QRC	2,530	89.78 ± 0.51
Full FitQT (Multi-Head + QFFN + QRC)	2595	95.17 ± 0.71

Table 8. Performance and efficiency comparison of models on the UIT-VSFC dataset. The Vietnamese large language models used as baselines include PhoBERT, ViBERT, and XLM-RoBERTa. Notably, the study focuses on the number of trainable parameters and the total training time to evaluate the practical efficiency of the proposed FitQT model.

Model	Test accuracy (%)	Test F1-score (%)	Trainable parameters	Total training time (min)	Avg time per Epoch
PhoBERT [76]	93.43	93.18	135000579	25.25	5.05 min
ViBERT [77]	91.76	91.28	177855747	30.40	5.07 min
XLM-RoBERTa [78]	93.46	93.05	278045955	25.77	5.15 min
FitQT (ours)	**88.72**	**88.02**	**3797**	**2.78**	**2.78 s**

4.7. Ablation study on the IMDb dataset

Table 7 presents the ablation study results on the IMDb dataset. To better understand the role of each architectural component, we conduct an ablation study on the IMDb dataset. For fairness and stability analysis, each model variant is trained and evaluated over nine independent runs. The reported results correspond to the mean accuracy and standard deviation across these runs.

The Quantum Single-Head Attention model achieves 76.61% ± 0.53, indicating that a basic quantum attention mechanism can already capture contextual information to a certain extent. Adding the Quantum Feed-Forward Network increases performance to 81.94% ± 0.61, showing that additional non-linear transformation substantially improves representation quality.

Replacing the single-head design with Quantum Multi-Head Attention further improves accuracy to 86.22% ± 1.00. This confirms that multiple heads allow the model to capture richer and more diverse token interactions. When Multi-Head Attention is combined with QFFN, performance increases significantly to 92.83% ± 0.56, highlighting the strong complementarity between contextual aggregation and feature refinement.

Introducing Quantum Residual Connections together with Multi-Head Attention results in 89.78% ± 0.51. While residual connections alone provide moderate gains, their contribution becomes more evident in the fully integrated model. The complete FitQT architecture achieves the highest performance at 95.17% ± 0.71.

Overall, the ablation results show that each component contributes meaningfully, but the strongest and most stable performance emerges when all modules are integrated within a unified architecture.

4.8. Results on the UIT-VSFC dataset

Table 8 presents a comparative analysis of performance and efficiency across models on the UIT-VSFC dataset. While large Vietnamese language models such as XLM-RoBERTa, PhoBERT, and ViBERT achieved high test accuracy and F1-scores (above 91%), they require a substantial number of trainable parameters (ranging from 135 million to 278 million) and training times of over 25 minutes. In contrast, the FitQT, despite slightly lower performance (88.72% accuracy and 88.02% F1-score), demonstrates a remarkable advantage in both training time and model compactness. It completes training in only 2.78 minutes and requires merely 3,797 trainable parameters, making it extremely lightweight compared to the baselines. Furthermore, its average training time per epoch is just 2.78 s, which highlights its potential for efficient deployment in resource-constrained environments or scenarios where rapid training and low memory usage are critical.

Importantly, all experiments were conducted under the same hardware configuration to ensure fairness and consistency in the evaluation. This uniform setup allows for a reliable comparison of computational efficiency, confirming that the advantages of the FitQT model are not due to differences in infrastructure but stem directly from its lightweight and optimized design.

On the UIT-VSFC dataset, comparisons focus on classical baselines, as prior quantum models lack publicly available implementations supporting Vietnamese data. Consequently, conclusions are restricted to assessing FitQT's competitiveness against classical architectures on this dataset.

5. Conclusion

In this study, we addressed the significant gap between fragmented component-level advances in Quantum Natural Language Processing (QNLP) and the pressing need for a unified, trainable Quantum Transformer architecture. We proposed FitQT, a novel and fully integrated FitQT model designed by integrating three key components: (1) a Gaussian Projected Quantum Multi-head Attention (GPMHA) mechanism, (2) a Quantum Feed-Forward Network (QFFN), and (3) most importantly, quantum residual connections to ensure stable training and effective information propagation across deep quantum circuits. Our primary objective was to construct a functionally complete and architecturally coherent model capable of tackling real-world NLP tasks with high efficiency.

Our empirical evaluations, conducted on both English and Vietnamese sentiment analysis datasets, demonstrate the effectiveness of the FitQT. On benchmark English datasets (Yelp, IMDb, Amazon), FitQT consistently outperformed existing quantum-based architectures, achieving a strong task-level accuracy of 95.17% on the IMDb dataset. We acknowledge that FitQT introduces a larger parameter count compared to several existing quantum baselines; therefore, part of the observed performance improvement may be attributed to increased model capacity rather than architectural design alone. A more controlled parameter-matched comparison will be explored in future work to further isolate the architectural contribution of FitQT. FitQT achieves competitive task-level performance relative to large-scale pretrained Transformer models such as BERT-Base and DistilBERT, despite being trained from scratch and using substantially fewer parameters. This consequently leads to a markedly faster training process. Moreover, the FitQT's strong performance on the Vietnamese UIT-VSFC dataset underscores the architecture's versatility and its potential for broader linguistic applications. We note, however, that direct comparisons with certain prior quantum architectures on the Vietnamese dataset were not conducted due to the absence of publicly available reproducible implementations supporting this language. Consequently, conclusions on the UIT-VSFC benchmark are limited to assessing FitQT's competitiveness against classical transformer models rather than establishing comprehensive quantum-to-quantum superiority across languages.

Future work will proceed on two primary directions. The immediate priority is to deploy and benchmark this hybrid model on Physical Noisy Intermediate-Scale Quantum hardware to evaluate its resilience to noise and its practical feasibility. Building upon this, our long-term vision is to address the hybrid architectural constraints, where quantum and classical components are still interdependent, by designing and formalizing an end-to-end quantum Transformer architecture. This ambitious effort will involve developing novel quantum subroutines for every constituent part—including multi-head combination, layer normalization, and the final prediction layer, aiming to minimize the classical, quantum interface and maintain quantum coherence throughout the entire computational pipeline. Such a fully quantum model could enable further investigation into whether quantum effects may offer empirical benefits for specific language tasks. Further explorations will also include scaling the model to a larger number of qubits and adapting it for more sophisticated applications such as question answering and machine translation.

Conflict of interest

The authors declare no conflict of interest.

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Author contributions

Phan Cong Lenh Pham: Conceptualization, Methodology, Experiments, Writing—original draft.
The Anh Le: Supervision, Review & Editing, Methodology.
All authors reviewed and approved the manuscript.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.24432/C57604 [79].

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